

Elementary Problems

Circle

Solutions

Version 1.0

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Basic Work

1. If $ABCD$ is cyclic, then

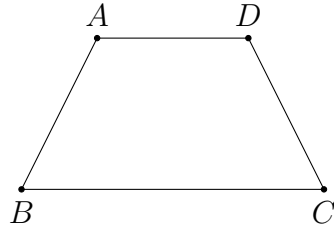
$$\angle B = 180^\circ - \angle D = \angle C$$

since $AD \parallel BC$. This implies $ABCD$ is an isosceles trapezoid.

Conversely, if $ABCD$ is an isosceles trapezoid, then

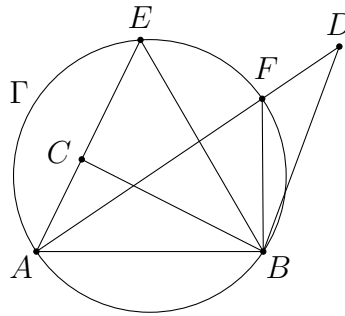
$$\angle B = \angle C = 180^\circ - \angle D.$$

Hence, $ABCD$ is cyclic.



2. Let AC and AD meet Γ again at E and F respectively. Then C lies between A and E , while F lies between A and D . Therefore, we have

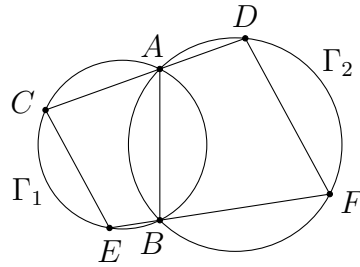
$$\angle ACB > \angle AEB = \angle AFB > \angle ADB.$$



3. We have

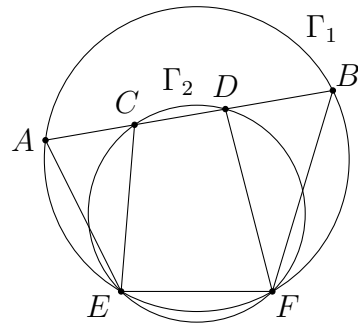
$$\angle FEC = \angle BEC = \angle BAD = 180^\circ - \angle DFB = 180^\circ - \angle DFE.$$

This implies $CE \parallel DF$.

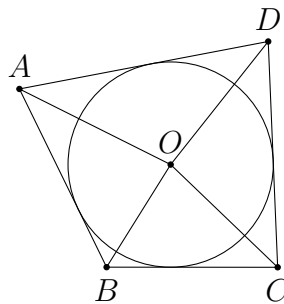


4. We have

$$\begin{aligned} \angle CEA &= \angle FEA - \angle FEC = (180^\circ - \angle ABF) - (180^\circ - \angle CDF) \\ &= \angle CDF - \angle ABF = \angle BFD. \end{aligned}$$



5. Since the sides of the quadrilateral are tangent to the circle, OA, OB, OC, OD are the internal angle bisectors of the interior angles.



It follows that

$$\begin{aligned}
\angle AOB + \angle COD &= (180^\circ - \angle BAO - \angle OBA) + (180^\circ - \angle DCO - \angle ODC) \\
&= 360^\circ - \frac{\angle BAD}{2} - \frac{\angle CBA}{2} - \frac{\angle DCB}{2} - \frac{\angle ADC}{2} \\
&= 360^\circ - \frac{360^\circ}{2} = 180^\circ.
\end{aligned}$$

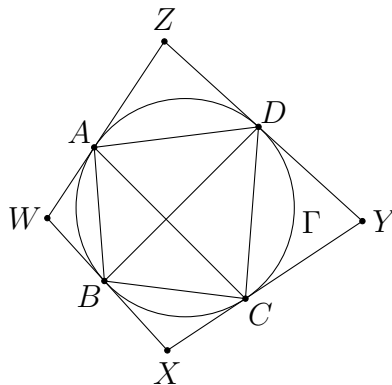
6. Since WA and WB are tangents to Γ , we have

$$\angle WAB = \angle ABW = \angle ACB.$$

Similarly, we have $\angle YCD = \angle CDY = \angle CBD$. Thus,

$$\begin{aligned}
\angle XWZ + \angle ZYX &= 180^\circ - \angle WAB - \angle ABW + 180^\circ - \angle YCD - \angle CDY \\
&= 360^\circ - 2\angle ACB - 2\angle CBD \\
&= 360^\circ - 2(\angle ACB + \angle CBD) \\
&= 360^\circ - 2 \times 90^\circ = 180^\circ.
\end{aligned}$$

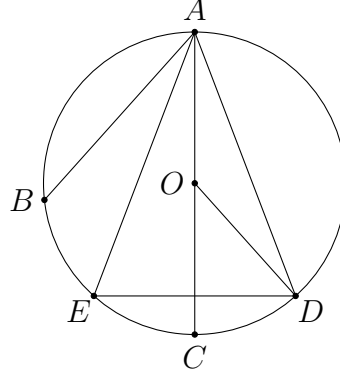
This implies $WXYZ$ is cyclic.



7. Note that

$$\angle EAC = \frac{1}{2}\angle BAC = \frac{1}{2}\angle COD = \angle CAD.$$

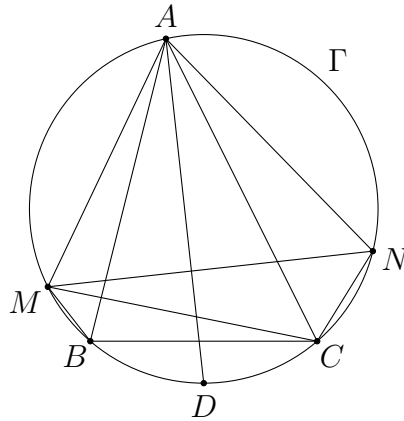
As AC is a diameter of the circle, this shows D and E are symmetric with respect to the line AC . Hence, $AC \perp DE$.



8. Let the internal angle bisector of $\angle BAC$ meet Γ again at D . Note that D is the midpoint of the arc BC . We find that

$$\begin{aligned} \angle ANM + \angle DAN &= \frac{1}{2}\angle ANC + (\angle BAN - \angle BAD) \\ &= \frac{1}{2}\angle ANC + \angle BMN - \angle BAD \\ &= \frac{1}{2}\angle ANC + \frac{1}{2}\angle BMA - \frac{1}{2}\angle BAC \\ &= \frac{1}{2}(\angle ANC + \angle BMA - \angle BMC) \\ &= \frac{1}{2}(\angle ANC + \angle CMA) = 90^\circ. \end{aligned}$$

This implies $MN \perp AD$.



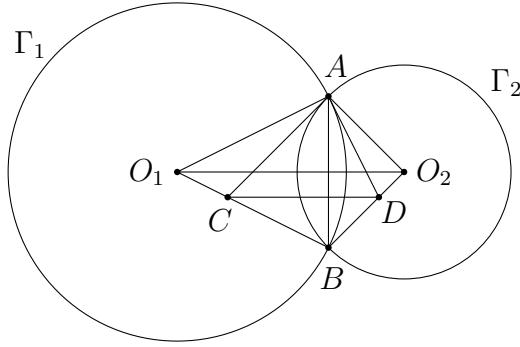
9. We have

$$\begin{aligned}\angle CAD &= \angle CAB + \angle BAD = (90^\circ - \angle BAO_2) + (90^\circ - \angle O_1AB) \\ &= 180^\circ - \angle O_1AO_2 = 180^\circ - \angle DBC\end{aligned}$$

due to symmetry. This implies A, C, B, D are concyclic. From this, we obtain

$$\angle BCD = \angle BAD = 90^\circ - \angle O_1AB = 90^\circ - \angle ABO_1 = \angle BO_1O_2.$$

Hence, $CD \parallel O_1O_2$.

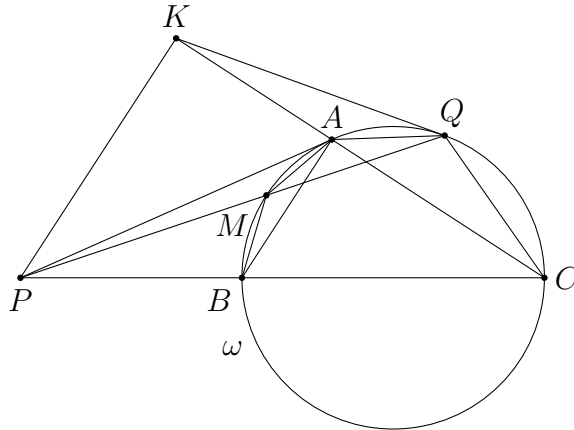


10. Since KQ is tangent to ω , we have $\triangle KAQ \sim \triangle KQC$. This yields

$$\frac{KA}{KQ} = \frac{KQ}{KC} = \frac{AQ}{QC}.$$

Hence, we have

$$\frac{KA}{KC} = \frac{KA}{KQ} \times \frac{KQ}{KC} = \frac{AQ^2}{QC^2}. \quad (1)$$



Similarly, from $\triangle PBM \sim \triangle PQC$, we have

$$\frac{PB}{PQ} = \frac{PM}{PC} = \frac{BM}{QC},$$

and hence

$$\frac{PB}{PC} = \frac{PB}{PQ} \times \frac{PM}{PC} \times \frac{PQ}{PM} = \frac{BM^2}{QC^2} \times \frac{PQ}{PM}. \quad (2)$$

To show that $\angle PKC = 90^\circ$, we need to show $KP \parallel AB$ since $\angle BAC = 90^\circ$. Thus, it suffices to show $\frac{KA}{KC} = \frac{PB}{PC}$. Comparing (1) and (2), we need to prove

$$AQ^2 = BM^2 \times \frac{PQ}{PM}. \quad (3)$$

Now, as PA is tangent to ω , we have $QPA \sim \triangle APM$. This yields

$$\frac{QP}{AP} = \frac{QA}{AM} = \frac{PA}{PM}.$$

Therefore, we have

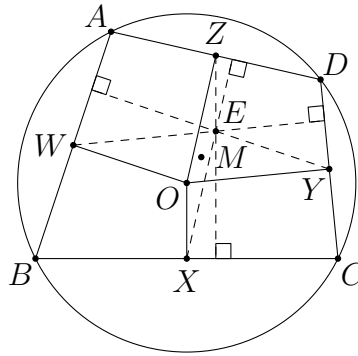
$$\frac{QA^2}{AM^2} = \frac{QP}{AP} \times \frac{PA}{PM} = \frac{QP}{PM}.$$

By noting $AM = BM$, we have established (3) as desired. This implies $\angle PKC = 90^\circ$ as discussed.

11. Let the cyclic quadrilateral be $ABCD$ and let O be the centre of $(ABCD)$. Let W, X, Y, Z be the midpoints of AB, BC, CD, DA respectively.

By the midpoint theorem, we have $WX \parallel AC$ and $ZY \parallel AC$. Thus, $WX \parallel ZY$. Similarly, $WZ \parallel XY$. This shows $WXYZ$ is a parallelogram. The diagonals WY and XZ bisect each other at a point M .

Let E be the reflection of O in the point M . Then $WOYE$ and $ZOXE$ are parallelograms since their diagonals bisect each other. As $OY \perp CD$, we have $WE \perp CD$. Similarly, we have $XE \perp DA$, $YE \perp AB$ and $ZE \perp BC$. This shows the four perpendiculars are concurrent at E .

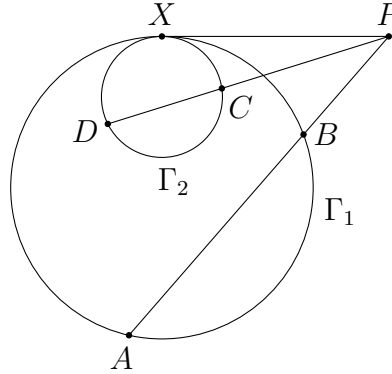


Power

12. We have

$$PA \times PB = PX^2 = PC \times PD.$$

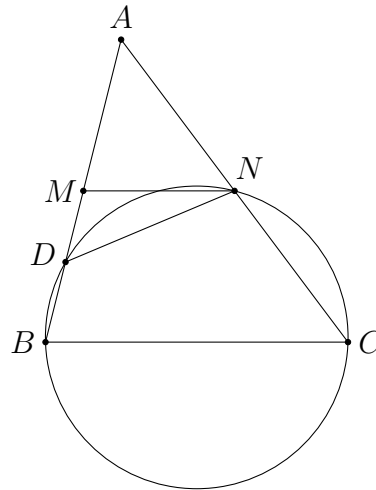
This shows A, B, C, D are concyclic.



13. Since D, B, C, N are concyclic, we have $AD \times AB = AN \times AC$. It follows that

$$AM \times AD = \left(\frac{1}{2}AB\right) \times AD = AN \times \left(\frac{1}{2}AC\right) = AN^2.$$

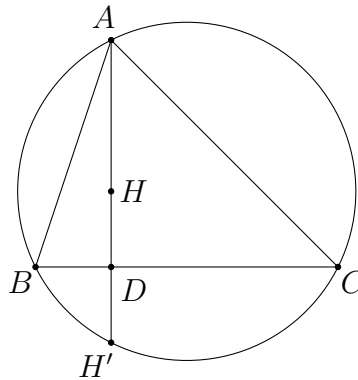
Hence, AN is tangent to (MDN) .



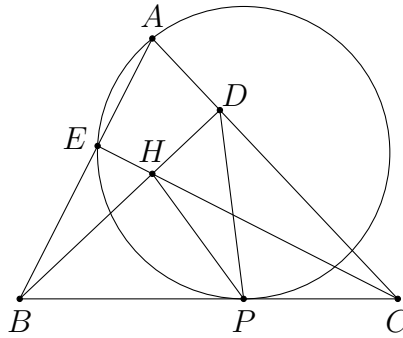
14. Let H' be the reflection of H in BC . Then

$$DB \times DC = DA \times DH'.$$

Since D lies on the segments BC and AH' , this implies A, B, H', C are concyclic. By the property of the orthocentre, H' is the reflection of the orthocentre of $\triangle ABC$ in BC . Hence, H is the orthocentre.



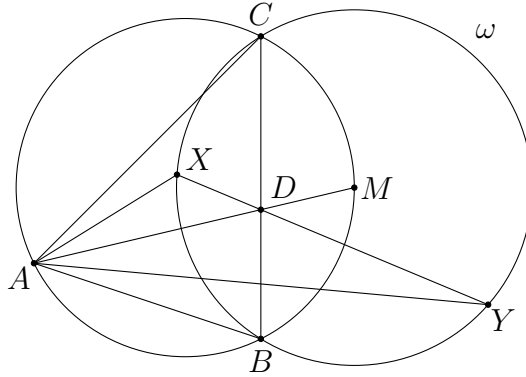
15. Note that A, E, H, D are concyclic. This implies $BH \times BD = BE \times BA$. Also, we have $BE \times BA = BP^2$. This yields $BH \times BD = BP^2$. Therefore, BC is tangent to (HPD) . It follows that $\angle CPD = \angle PHD$.



16. Note that C lies on ω as $MB = MC$. We have

$$DX \times DY = DB \times DC = DA \times DM.$$

This implies A, Y, M, X are concyclic. Together with $MX = MY$, we know that AM bisects $\angle YAX$.



17. Let Γ_1 be the circle with diameter AB and Γ_2 the circle with diameter AC . Let H be the orthocentre of $\triangle ABC$. Then we have

$$HM \times HN = HB \times HB'$$

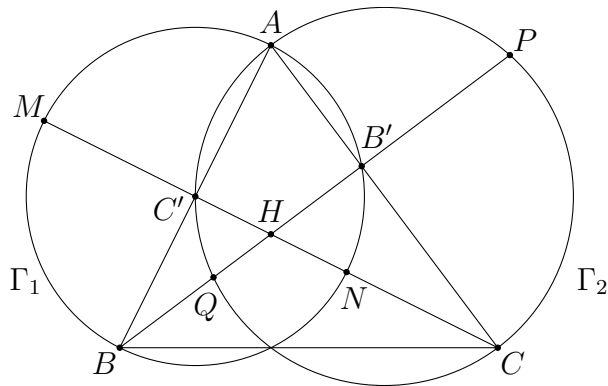
and

$$HP \times HQ = HC \times HC'.$$

Note that B, C, B', C' are concyclic. Therefore, $HB \times HB' = HC \times HC'$. It follows that

$$HM \times HN = HP \times HQ.$$

Hence, M, N, P, Q are concyclic.



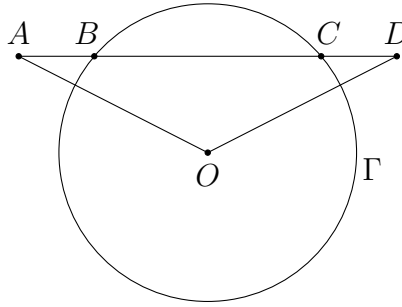
18. Note that $OA = OD$ if and only if the powers of A and D with respect to Γ are equal, i.e $AB \times AC = DB \times DC$. We have

$$AB \times AC = AB(AB + BC)$$

and

$$DB \times DC = CD(CD + BC).$$

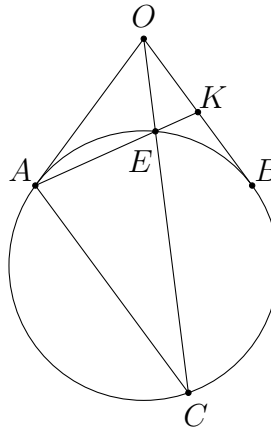
Since the function $f(x) = x(x + BC)$ is increasing for positive x , $AB \times AC = DB \times DC$ holds if and only if $AB = CD$.



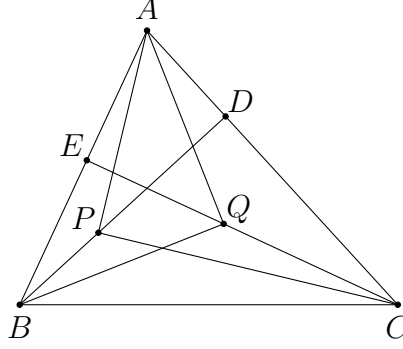
19. Note that

$$\angle EOK = \angle COK = \angle OCA = \angle ECA = \angle EAO.$$

This shows KO is a tangent to (AEO) . Thus, we have $KO^2 = KA \times KE$. Together with $KB^2 = KA \times KE$, we get $KO = KB$. Thus, K is the midpoint of OB .



20. Since $\angle CPA = \angle ADP = 90^\circ$, we have $AP^2 = AD \times AC$. Due to symmetry, we have $AQ^2 = AE \times AB$. Note that B, C, D, E are concyclic. Thus, $AE \times AB = AD \times AC$. Therefore, we get $AP^2 = AQ^2$, and hence $AP = AQ$.



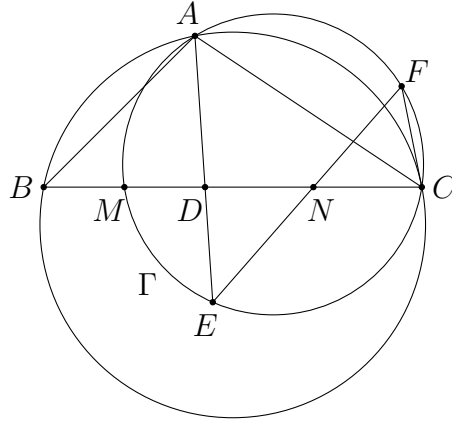
21. Since A, M, E, C are concyclic, we have $AD \times ED = MD \times CD$. Using the midpoints, we find that

$$AD \times ED = MD \times CD = \left(\frac{1}{2}BD\right)(2ND) = BD \times ND.$$

This implies A, B, E, N are concyclic. It follows that

$$\angle FCA = \angle FEA = \angle NEA = \angle NBA = \angle CBA.$$

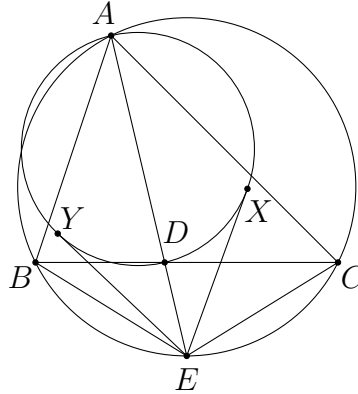
Hence, FC is tangent to (ABC) .



22. From $\angle EBD = \angle EAC = \angle BAE$, we have $\triangle BDE \sim \triangle ABE$. This yields

$$EB^2 = ED \times EA.$$

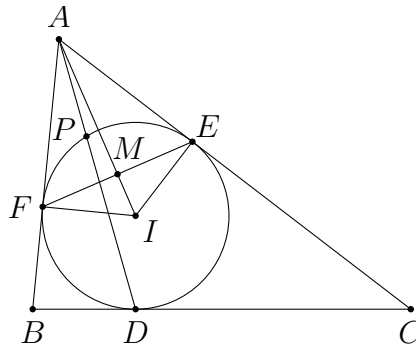
Also, we have $EX^2 = EY^2 = ED \times EA$. This implies $EX = EY = EB$, which is also equal to EC . Therefore, B, C, X, Y are concyclic, lying on a circle with centre E .



23. Note that AI is the perpendicular bisector of EF . Thus, M lies on AI and also $\angle EMA = 90^\circ$. Since we also have $\angle AEI = 90^\circ$, we get

$$AM \times AI = AE^2.$$

In addition, we have $AE^2 = AP \times AD$ by considering the power of A with respect to the incircle. Hence, $AM \times AI = AP \times AD$. This shows P, D, I, M are concyclic.



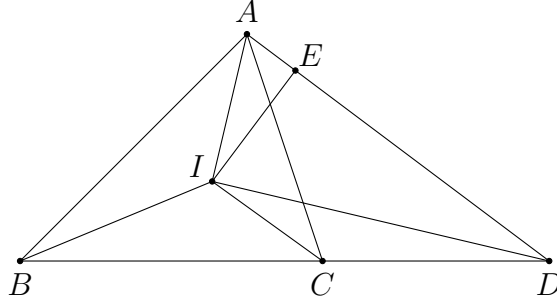
24. Firstly, from

$$\angle CID = \angle CIA - 90^\circ = \frac{B}{2} = \angle CBI,$$

we find that DI is tangent to (IBC) . This implies $DI^2 = DB \times DC$.

Next, from $\angle DIA = \angle IED = 90^\circ$, we have $DI^2 = DE \times DA$.

Therefore, we find that $DB \times DC = DE \times DA$. This shows A, B, C, E are concyclic.



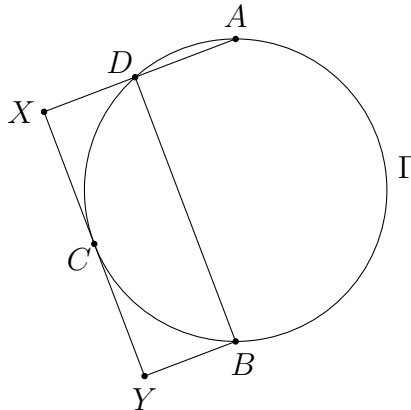
25. Let AX meet Γ again at D . WLOG assume D lies between A and X . Note that $DX Y B$ is a rectangle since $\angle BDA = \angle BYX = \angle YXD = 90^\circ$. Hence,

$$AX \cdot BY = AX \cdot DX = XC^2.$$

Note that the line joining C and the centre of Γ is perpendicular to XY , and hence to BD . Therefore, C is the midpoint of the arc BD , and hence the midpoint of XY . Therefore,

$$AX \cdot BY = XC^2 = \left(\frac{1}{2}XY\right)^2 = \frac{1}{4}XY^2.$$

The result then follows.

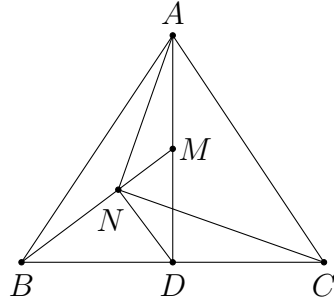


26. As $\angle MDB = \angle DNM = 90^\circ$, we have $MD^2 = MN \times MB$ and $\angle MDN = \angle DBM$. From $MA = MD$, we get $MA^2 = MD^2 = MN \times MB$. Thus, $\angle NAM = \angle MBA$.

Now, we have

$$\angle ACD = \angle CBA = \angle DBM + \angle MBA = \angle MDN + \angle NAM = 180^\circ - \angle DNA.$$

This shows A, N, D, C are concyclic, and hence $\angle CNA = \angle CDA = 90^\circ$.



27. Let D and E be the feet of the altitudes from B and C respectively. As

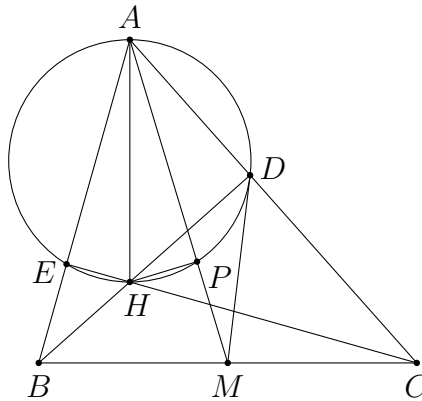
$$\angle ADH = \angle AEH = \angle APH = 90^\circ,$$

points A, E, H, P, D are concyclic. Recall that $MB = ME = MD = MC$. Thus, we have

$$\angle HDM = \angle MBD = 90^\circ - C = \angle HAD.$$

This shows MD is tangent to (AED) . Hence,

$$MP \times MA = MD^2 = MB^2.$$



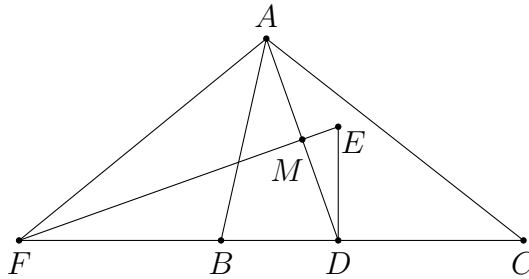
28. Let EM meet BC at F . Note that FME is the perpendicular bisector of AD . From $\angle FMD = \angle EDF = 90^\circ$, we have $FM \times FE = FD^2 = FA^2$.

Next, we find that

$$\angle FAB = \angle FAD - \angle BAD = \angle ADF - \angle DAC = \angle ACB.$$

This shows AF is tangent to (ABC) . Hence, we have $FA^2 = FB \times FC$.

Combining all these, we get $FM \times FE = FB \times FC$. This implies E, M, B, C are concyclic.



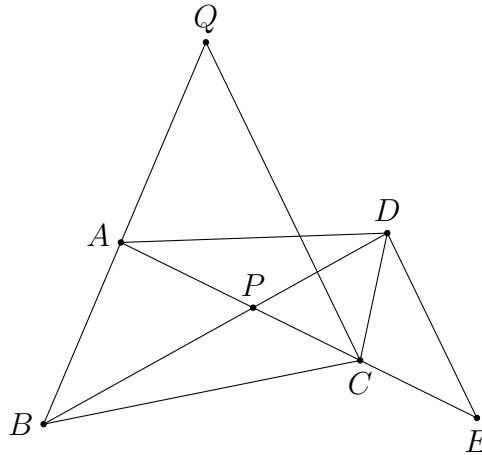
29. Let E be the point on the ray AC such that $CE = CD$. Then the condition becomes

$$PA \times PE = PA(PC + CE) = PB \times PD.$$

This implies A, B, E, D are concyclic. It follows that

$$\angle DBQ = \angle DEA = \frac{1}{2}\angle DCA = \angle DCQ.$$

Hence, B, C, D, Q are concyclic.



30. Let P be the point on the side BC such that A, D, P, C are concyclic. Then we have

$$BD \cdot BA = BP \cdot BC.$$

Using the condition $BC^2 = BD \cdot BA + CE \cdot CA$, we get

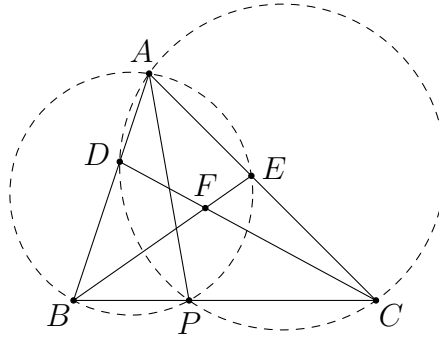
$$BC^2 - BP \cdot BC = CE \cdot CA,$$

i.e. $CP \cdot CB = CE \cdot CA$. This implies A, E, P, B are concyclic.

From the concyclic points, we deduce that

$$\angle FDA + \angle AEF = \angle CPA + \angle APB = 180^\circ.$$

Hence, A, D, F, E are concyclic.



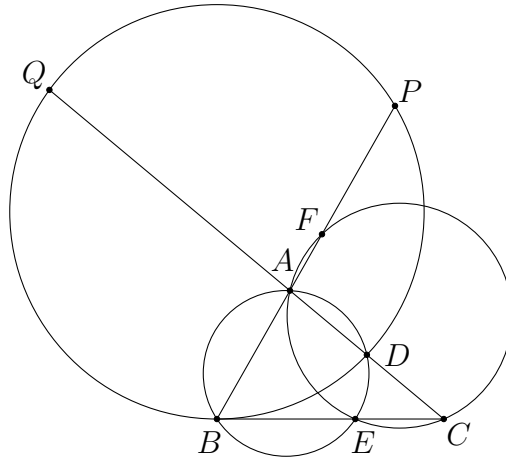
31. Let the line AC meet (BDP) again at Q . From the power of points theorem, we have

$$BE \times BC = BA \times BF, \quad (1)$$

$$CE \times CB = CD \times CA, \quad (2)$$

$$CB^2 = CD \times CQ, \quad (3)$$

$$AD \times AQ = AB \times AP. \quad (4)$$



Adding (1) and (2), we get

$$CB^2 = BC(BE + EC) = BA \times BF + CD \times CA. \quad (5)$$

Comparing (3) and (5), we obtain

$$CD \times CQ = BA \times BF + CD \times CA.$$

This implies

$$CD \times AQ = BA \times BF. \quad (6)$$

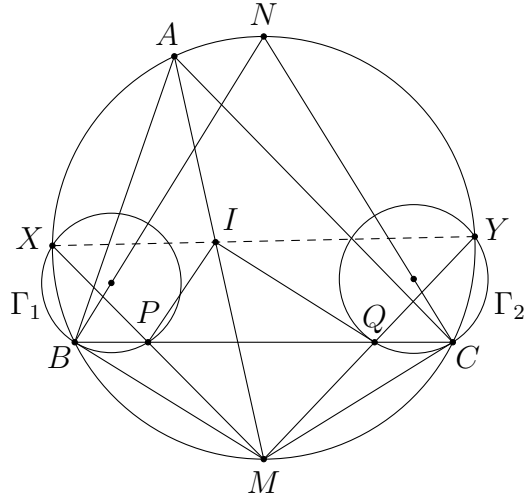
From (4) and (6) and noting $AD = CD$, we find that $AP = BF$. Thus, $AB = FP$.

32. (It is not hard to prove that X and Y lie on (ABC) , but this is not needed in the following proof.)

Note that MN is a diameter of (ABC) . So $\angle MBN = 90^\circ$. As the centre of (XBP) lies on BN , BM is a tangent of (XBP) . Hence, we get

$$MP \times MX = MB^2 = MI^2.$$

The last equality holds since $MB = MI$ as a well-known property of the incentre. From this, we obtain $\triangle MPI \sim \triangle MIX$. Thus, $\angle MPI = \angle MIX$. Similarly, we have $\angle MQI = \angle MIY$. Then, X, I, Y are collinear if and only if $\angle MIX + \angle MIY = 180^\circ$, if and only if $\angle MPI + \angle MQI = 180^\circ$. This is equivalent to P, M, Q, I being concyclic.



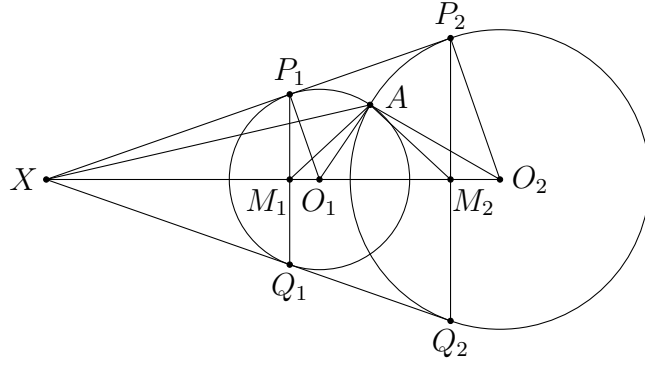
33. Let X be the external homothetic centre of C_1 and C_2 . Thus, X lies on P_1P_2 and O_1O_2 . Since $\angle XP_1O_1 = \angle O_1M_1P_1 = 90^\circ$, we have

$$O_1M_1 \times O_1X = O_1P_1^2.$$

As $O_1A = O_1P_1$ is the radius of C_1 , we have $O_1M_1 \times O_1X = O_1A^2$. This shows O_1A is a tangent to (XM_1A) and hence $\angle M_1AO_1 = \angle M_1XA$.

Similarly, we have $\angle M_2AO_2 = \angle M_2XA$. This yields $\angle M_1AO_1 = \angle M_2AO_2$ so that

$$\angle O_1AO_2 = \angle O_1AM_2 + \angle M_2AO_2 = \angle O_1AM_2 + \angle M_1AO_1 = \angle M_1AM_2.$$

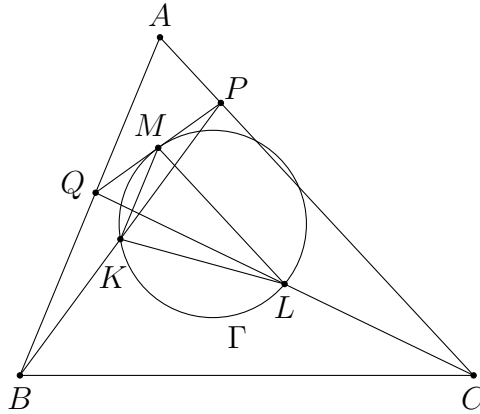


34. By the midpoint theorem, we have $ML \parallel PC$. Thus,

$$\angle APQ = \angle LMP = \angle MKL.$$

Similarly, we have $\angle AQP = \angle MLK$. Thus, $\triangle APQ \sim \triangle MKL$. This implies

$$\frac{AP}{MK} = \frac{AQ}{ML}.$$



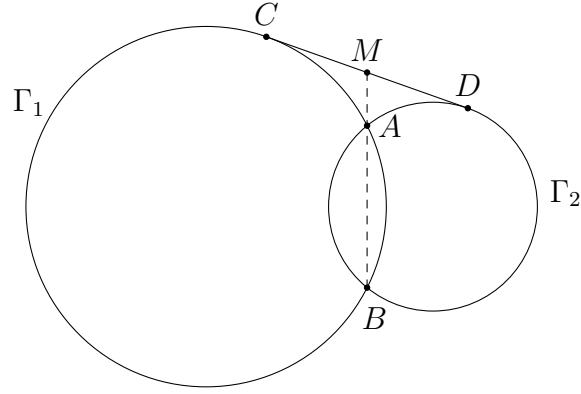
By the midpoint theorem, we have $ML = \frac{1}{2}PC$ and similarly $MK = \frac{1}{2}QB$. Therefore, we obtain

$$QA \times QB = QA \times (2MK) = PA \times (2ML) = PA \times PC.$$

As P and Q lie inside (ABC) , this shows the powers of P and Q with respect to (ABC) are equal. Hence, $OP = OQ$.

Radical Axis

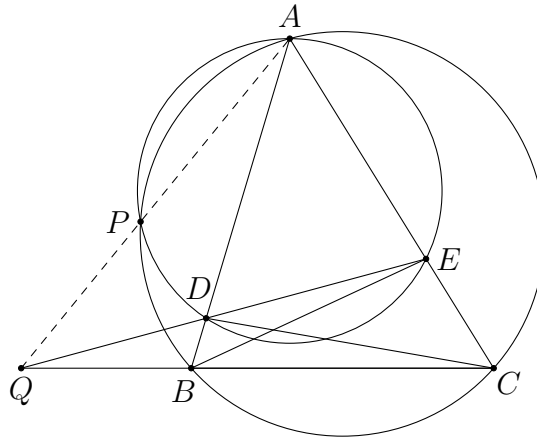
35. Let M be the midpoint of CD . Then the power of M with respect to Γ_1 is MC^2 , while that with respect to Γ_2 is MD^2 . These are equal. Thus, M lies on the radical axis AB of Γ_1 and Γ_2 . In other words, AB bisects CD .



36. Since $\angle ECD = \angle DAC = \angle EBD$, points B, C, E, D are concyclic. This yields

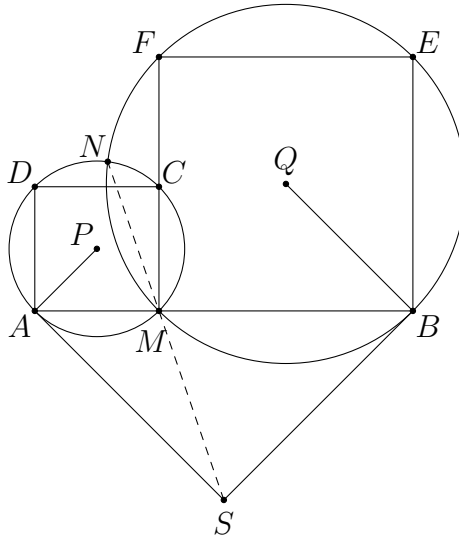
$$QB \times QC = QD \times QE.$$

Thus, the powers of Q with respect to (ABC) and (ADE) are equal. Therefore, Q lies on the radical axis AP of these circles.



37. Let the line through A perpendicular to AP meet the line through B perpendicular to BQ at S . Since $\angle BAP = \angle QBA = 45^\circ$, we have $\angle SAB = \angle ABS = 45^\circ$. This implies $SA = SB$ and S is a fixed point independent of M .

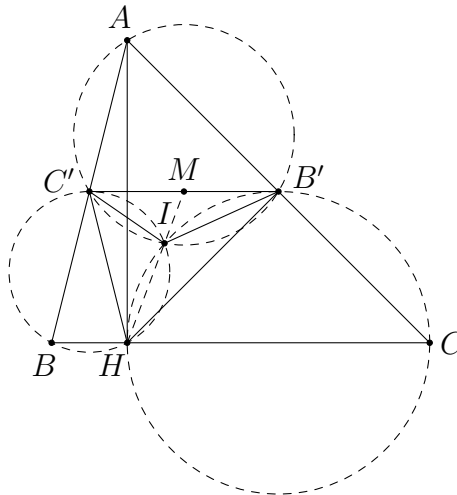
Note that the line MN is the radical axis of the two circles. As $\angle SAP = \angle QBS = 90^\circ$, SA and SB are tangents to the two circles respectively. As $SA = SB$, the powers of S with respect to the two circles are equal. Hence, the fixed point S lies on the radical axis MN .



38. Let I be the intersection of $(C'BH)$ and $(CB'H)$. Then

$$\angle IC'A = \angle IHB = \angle IB'C.$$

This implies I lies on $(AC'B')$. Indeed, I is the Miquel point of $\triangle HB'C'$ with respect to $\triangle ABC$.



Next, since $\angle AHB = 90^\circ$ and C' is the midpoint of AB , we have $C'H = C'B$. By the midpoint theorem, we have $C'B' \parallel BC$. Thus,

$$\angle HC'B' = \angle C'HB = \angle HBC'.$$

This shows $B'C'$ is a tangent to $(C'BH)$. Similarly, it is also a tangent to $(CB'H)$. Let M be the midpoint of $C'B'$. Then the powers of M with respect to $(C'BH)$ and $(CB'H)$ are equal as $MC'^2 = MB'^2$. Thus, M lies on the radical axis of the two circles, which is the line HI since the two circles intersect at H and I .

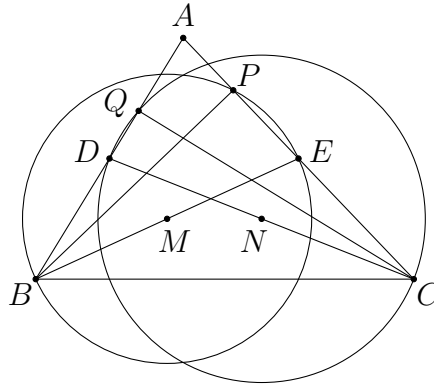
39. Let P and Q be the feet of the altitudes from B and C respectively. Note that P lies on the circle with diameter BE , and Q lies on the circle with diameter CD .

The powers of A with respect to (BEP) and (QDC) are $AP \times AE$ and $AQ \times AD$ respectively. These are equal since

$$AP \times AE = \frac{1}{2} \times AP \times AC = \frac{1}{2} \times AQ \times AB = AQ \times AD.$$

Note that we have used the fact that B, C, P, Q are concyclic.

Therefore, A lies on the radical axis of the two circles. Let M and N be the midpoints of BE and CD respectively. Then M and N are the centres of (BEP) and (QDC) respectively. By the midpoint theorem, we have $DE \parallel BC$. Hence, $MN \parallel BC$. Thus, the A -altitude is perpendicular to MN . This implies the A -altitude is the radical axis of the two circles.



40. Let the circle Γ_1 with diameter AC meet CD again at P . Let the circle Γ_2 with diameter BD meet AB again at Q .

Note that $\angle APC = 90^\circ$. Therefore, the power of E with respect to Γ_1 is

$$EP \times EC = EA \cos \angle AEC \times EC.$$

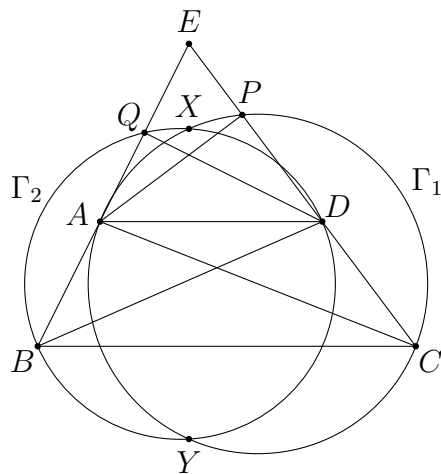
Similarly, the power of E with respect to Γ_2 is

$$ED \cos \angle BED \times EB.$$

Since $AD \parallel BC$, we have $\triangle EAD \sim \triangle EBC$. This implies $\frac{EA}{EB} = \frac{ED}{EC}$. Therefore,

$$EA \times EC \cos \angle AEC = ED \times EB \cos \angle BED.$$

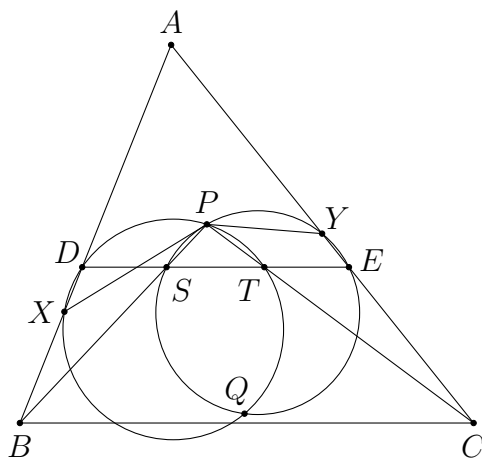
Thus, the powers of E with respect to the two circles are equal. Hence, E lies on the radical axis XY of Γ_1 and Γ_2 .



41. Let (PDT) meet AB again at X . Let (PSE) meet AC again at Y . Suppose X lies between B and D (the other cases can be handled similarly). Then we have

$$\angle XPC = \angle XPT = \angle XDT = 180^\circ - \angle CBX.$$

This implies X lies on (PBC) . Similarly, Y lies on (PBC) . Thus, X, B, C, Y, P are concyclic.



The power of A with respect to (PDT) is $AD \times AX$, while that with respect to (PSE) is $AE \times AY$. As $DE \parallel BC$, we have $\frac{AD}{AE} = \frac{AB}{AC}$. Hence,

$$AD \times AX = AB \times AX \times \frac{AE}{AC} = AC \times AY \times \frac{AE}{AC} = AE \times AY.$$

Therefore, A lies on the radical axis PQ of (PDT) and (PSE) .

42. Let Y be the intersection of BQ and CP . We need to show A, X, Y are collinear.

Let DE meet BQ at F . Firstly, since $DE \parallel BC$ by the midpoint theorem, we have

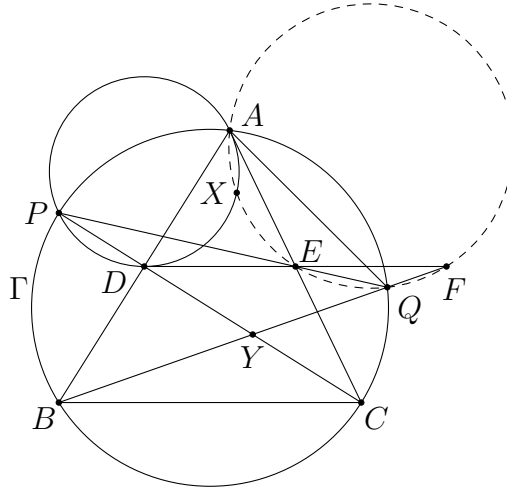
$$\angle FQA = 180^\circ - \angle AQB = 180^\circ - \angle ACB = 180^\circ - \angle AED = \angle FEA.$$

This implies F lies on (AEQ) .

Secondly, from

$$\angle FQP = 180^\circ - \angle PQY = 180^\circ - \angle PCB = 180^\circ - \angle CDE = \angle FDP,$$

points P, D, Q, F are concyclic. Therefore, we have $YD \times YP = YQ \times YF$. This implies Y lies on the radical axis AX of (APD) and (AEQ) . It follows that BQ, CP, AX are concurrent.



43. (a) If $AB \parallel CD$, then each of ABX_2X_1 , $X_2X_1W_1W_2$, W_1W_2CD is an isosceles trapezoid. Hence, $ABCD$ is an isosceles trapezoid and A, B, C, D are concyclic. Otherwise, let AB meet CD at P . From

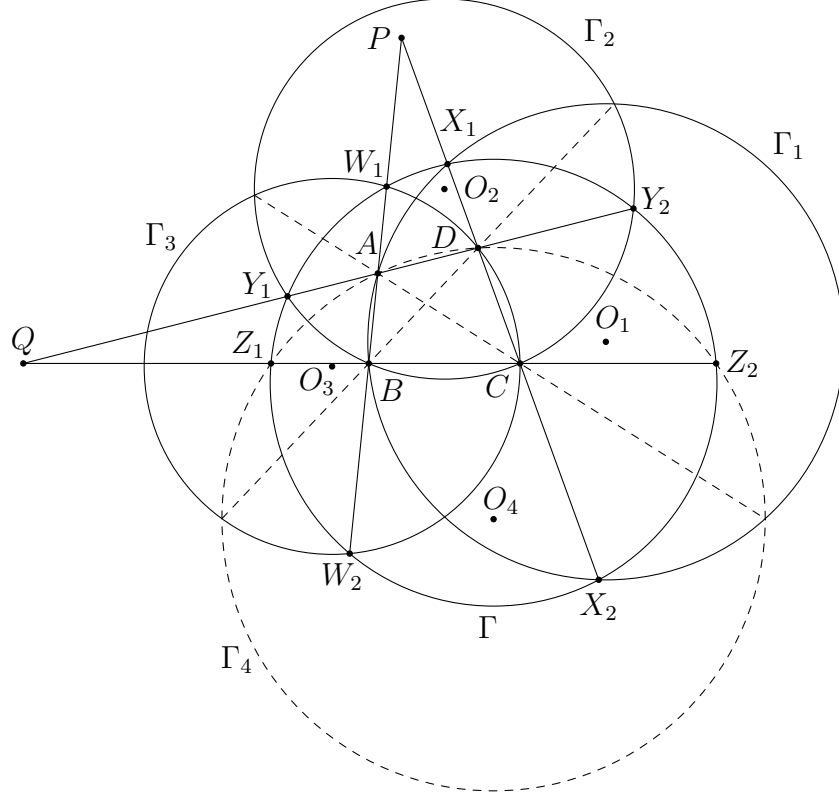
$$PA \times PB = PX_1 \times PX_2 = PW_1 \times PW_2 = PD \times PC,$$

points A, B, C, D are concyclic.

If $AD \parallel BC$, then each of $DABC$, BCY_2Y_1 , $Y_2Y_1Z_1Z_2$ is an isosceles trapezoid. Hence, DAZ_1Z_2 is an isosceles trapezoid and D, A, Z_1, Z_2 are concyclic. Otherwise, let AD meet BC at Q . From

$$QA \times QD = QB \times QC = QY_1 \times QY_2 = QZ_1 \times QZ_2,$$

points D, A, Z_1, Z_2 are concyclic.

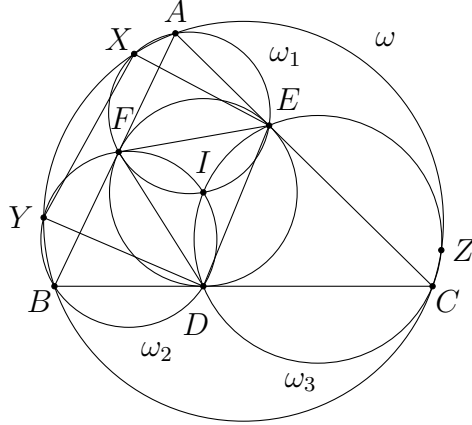


- (b) Let the centre of Γ_j be O_j for $j = 1, 2, 3, 4$. Note that A lies on Γ_1 and Γ_4 . From $CX_1 \times CX_2 = CZ_1 \times CZ_2$, C lies on the radical axis of Γ_1 and Γ_4 . Therefore, line AC is the radical axis of these two circles. In particular, $O_1O_4 \perp AC$. Similarly, we have $O_2O_3 \perp AC$ so that $O_1O_4 \parallel O_2O_3$. For similar reasons, we obtain $O_2O_1 \parallel O_3O_4$. This shows $O_1O_2O_3O_4$ is a parallelogram.

44. (a) We have

$$\begin{aligned} \angle YXE + \angle EDY &= (\angle YXA - \angle EXA) + (\angle EDF + \angle FDY) \\ &= (180^\circ - \angle ABY) - \angle EFA + \angle EDF + \angle FBY \\ &= 180^\circ - \angle EFA + \angle EDF = 180^\circ. \end{aligned}$$

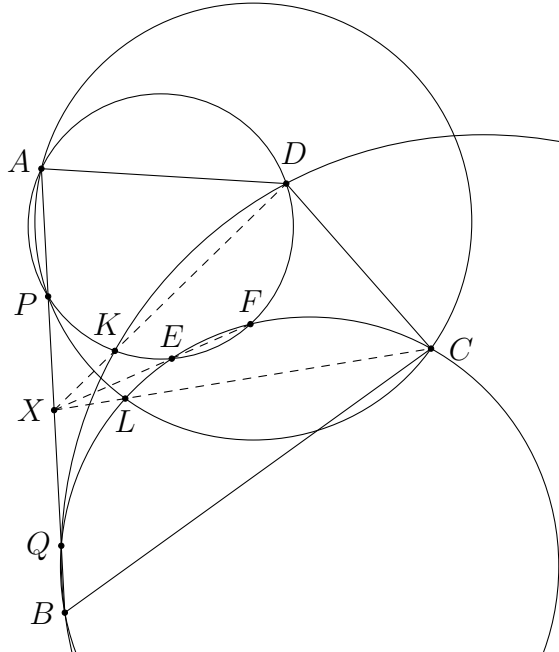
The last equality holds since AF is tangent to (DEF) . This implies X, Y, D, E are concyclic.



(b) Similarly, Y, Z, E, F are concyclic, Z, X, F, D are concyclic. The three groups of concyclic points determine three circles. The radical axes of the three pairs of circles are DX, EY, FZ respectively. Hence, they are concurrent at the radical centre of the three circles.

45. Let $(APKD)$ meet $(BCLQ)$ at E and F . Let EF meet AB at X .

Consider the circles $(APLC)$, $(APEF)$ and $(FELC)$. Their pairwise radical axes are AP, EF, LC respectively. Hence, AP, EF, LC are concurrent at the radical centre X of the circles.

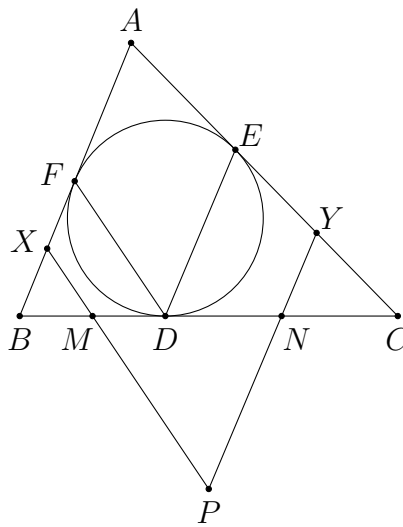


Similarly, by considering the radical centre of the circles $(DKQB)$, $(FEQB)$ and $(KEFD)$, we find that BQ, EF, KD are concurrent at X .

Therefore, we have $XK \times XD = XE \times XF = XL \times XC$. This implies D, K, L, C are concyclic.

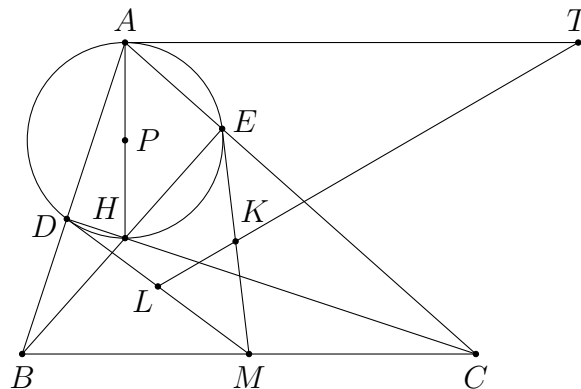
46. Let M and X be the midpoints of BD and BF respectively. Then ℓ_b is the line MX . Since $MB^2 = MD^2$, the powers of M with respect to the degenerated circle B and the incircle are equal. The same holds for the point X . Thus, ℓ_b is the radical axis of B and the incircle.

Similarly, ℓ_c is the radical axis of C and the incircle. Therefore, P is the radical centre of B, C and the incircle. It follows that $PB^2 = PC^2$, and hence $PB = PC$.



47. Let H be the orthocentre of $\triangle ABC$. Note that A, D, H, E are concyclic, lying on the circle with diameter AH .

Firstly, DM is tangent to (ADE) . Indeed, let P be the centre of (ADE) . Recall that PM is a diameter of the nine point circle of $\triangle ABC$, and D lies on this nine point circle. Thus, $\angle MDP = 90^\circ$. Similarly, EM is tangent to (ADE) .



As $LD^2 = LM^2$, the powers of L with respect to (ADE) and the degenerated circle M are equal. The same holds for the point K . Thus, LK is the radical axis of (ADE) and M . Hence, the powers of T with respect of the two circles are equal.

Since $AT \parallel BC$, $AT \perp AP$. This shows AT is tangent to (ADE) . It follows that the power of T with respect to (ADE) is TA^2 . Thus, $TA^2 = TM^2$, which implies $TA = TM$.

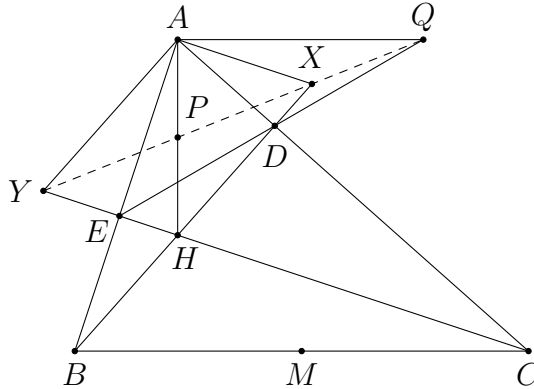
48. Let X be the point on the line BD such that $XA \perp AB$. Let Y be the point on the line CE such that $YA \perp AC$. From

$$\angle BAX = 90^\circ = \angle XDA,$$

we have $XA^2 = XB \times XD$. This shows the powers of X with respect to the degenerated circle A and $(BCDE)$ are equal. The same holds for the point Y due to symmetry. Thus, XY is the radical axis of A and $(BCDE)$.

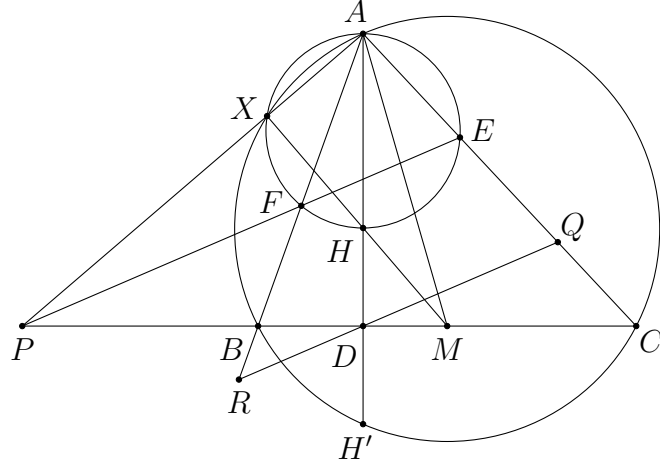
Note that $AYHX$ is a parallelogram. Therefore, the midpoint P of AH is the midpoint of XY . Thus, P lies on the radical axis of A and $(BCDE)$.

Next, as $\angle CAQ = \angle ACB = \angle DEA$, we have $QA^2 = QD \times QE$. This shows Q lies on the radical axis of A and $(BCDE)$. Hence, PQ is the radical axis. As M is the centre of $(BCDE)$, this implies $PQ \perp AM$.



49. We first recall a few well-known facts in triangle geometry. Let M be the midpoint of BC . Let the ray MH meet (ABC) at X . Then $\angle HXA = 90^\circ$. This is because the reflection of H in M coincides with the reflection of A in the circumcentre of $\triangle ABC$. Hence, X lies on $(AFHE)$.

The radical centre of $(AXFHE)$, $(BCEF)$ and $(AXBC)$ is the intersection of the lines AX, BC, EF , i.e. the point P . Therefore, P lies on AX . As $MH \perp AP$ and $AH \perp PM$, H is the orthocentre of $\triangle APM$.



We now return to the proof. Let H' be the reflection of H in BC . As H is the orthocentre of $\triangle APM$ as well as the orthocentre of $\triangle ABC$, H' lies on (APM) and (ABC) . Thus,

$$DP \times DM = DA \times DH' = DB \times DC.$$

Next, from $\angle RQC = \angle FEC = \angle RBC$, points B, R, C, Q are concyclic. Thus,

$$DB \times DC = DQ \times DR.$$

Therefore, we get $DP \times DM = DQ \times DR$. So P, R, M, Q are concyclic.